

Architect's Directive to Cursor Dev Agent – xChat Polish for HNWI Options Strategy Mastery

Hey Cursor team,

I am the Architect – 20+ years as a senior finance advisor and options trading specialist. I design, build, test, and deploy production-grade systems for HNWI clients who manage multiple portfolios across taxable, IRA, and trust accounts. Every feature I ship must deliver **conservative, balanced, or aggressive** outlooks with iron-clad risk management, tax awareness, and seamless integration into the live workspace (portfolios, accounts, positions, watchlists, balances).

We are polishing **xChat (advisor persona)** to become the premium, xAI-powered command center for portfolio-level options strategy construction. Leverage the existing Grok-style response modes (Auto / Expert / Heavy – visible in the Templates / composer dropdown) and the current **advisor persona** (see attached Edit xPersona screenshot). The goal is a guided, multi-account workflow that feels like sitting across from me in a high-net-worth strategy session.

1. Core Requirements – Grok-Style Modes + HNWI Restriction

- **Adapt & restrict the three modes** inside the **advisor persona** (System Prompt + Custom Instructions):
 - **Auto** → balanced, concise, production-ready recommendations.
 - **Expert** → deep quantitative analysis, Greeks, volatility surface, tax implications.
 - **Heavy** → full multi-agent orchestration, scenario modeling, backtest-style outputs, conservative stress tests.

- Update the **System Prompt** (already loaded in the screenshot):

text

****You are The Advisor**** – the trusted advisor for aTx Finance HNWI users. Calm, precise, long-term thinker focused exclusively on wealth preservation, income generation, and risk-adjusted growth across multiple portfolios/accounts.

You have direct access to the signed-in user's full ****aTx workspace**** (portfolio, watchlist, positions, balances) + live quotes, option chains, and market data via xAI tools.

Always operate in one of three modes the user selects (Auto / Expert / Heavy). Tailor depth accordingly while staying strictly within conservative/balanced/aggressive frameworks. Never speculate outside the user's actual holdings or risk tolerance.

- Add **Custom Instructions** block (Optional field in persona editor):
 - Enforce HNWI guardrails: account-type awareness (taxable vs tax-advantaged), position sizing $\leq 5\text{--}10\%$ of account equity (conservative default), margin rules, tax-lot matching, liquidity requirements.
 - Always cross-reference current portfolio holdings before suggesting new legs.

2. Top 5 Strategy Templates – Quick-Select Support

Add native support for the **Top 5 institutional-grade options strategies** (seed these into the Templates UI and as chat quick-actions):

1. **Wheel Strategy** (conservative income engine)
2. **Iron Condor** (range-bound neutral)
3. **Covered Call** (income overlay on long stock)
4. **Bull Put Spread / Credit Spread** (defined-risk bullish)
5. **Protective Put / Collar** (downside protection)

UI Placement:

- Enhance the **Templates** screen (see second screenshot) with a new section “**Options Portfolio Strategies**” containing 5 cards (icons + one-line description).
- In xChat composer, add persistent quick-chip buttons for these 5 templates + “Select Strategy Template”.

3. Guided Workflow When Template Is Selected

When user picks a template (chat command, button, or template card):

- **Step 1 – Portfolio/Account Discovery**

Use existing workspace tools to list **all accounts/portfolios** the user has access to. Present a clean summary table (Account Name | Type | Current Equity | Open Positions).

- **Step 2 – Per-Account Preference Capture (structured NL slots)**

For each account, launch a guided dialog (reuse/extend existing NL collect-inputs pattern from `xchat/nl-workflows/`):

- Risk outlook: Conservative / Balanced / Aggressive
- Capital allocation % for this strategy
- Time horizon (days/weeks/months to expiration)
- Primary goal: Income / Capital appreciation / Protection
- IV outlook preference (high / neutral / low)
- Any hard constraints (e.g., “do not sell covered calls on these 3 holdings”)
- Tax considerations flag

Store responses temporarily in session + persist as structured JSON in a new or extended `options_strategy_preferences` collection (per portfolio/account).

- **Step 3 – Async Multi-Agent Batch Job**

Once all preferences are captured (or user says “build now”):

- **Trigger an async Strategy Job** via the existing Spring `StrategyJobService` (`POST /api/strategy-jobs` with `jobType: "portfolio_strategy_build"`).
- **Multi-agent orchestration (async batch)** – align with Phase 1 xChat Hardcore (`atx-multi-agent.md`):
 - Master orchestrator agent
 - Sub-agents (one per account or per template leg): chain analysis, risk/greeks calc, position sizing, payoff modeling, tax impact, correlation check across portfolios.
 - Use existing RAG (`options-strategy/ collection + options-coreskills.md`) for canonical narrative + risk/outlook.
- Job runs in background (Redis hourly cap respected, ShedLock, audit trail).
- On completion: notify user in xChat thread with `correlationId`, save full artifact to `strategy_jobs` , and offer “Open in xStrategyBuilder / xOptions” deep link.

- **Step 4 – Deliver & Act**

Final response in xChat includes:

- Summary table per account
- Recommended legs (with live mid prices, Greeks, breakeven, max loss/profit)
- ApexCharts payoff diagram (reuse existing xOptions charting)
- One-click “Load into xOptions builder” or “Execute preview” (future IBKR path)
- Conservative/Balanced/Aggressive risk rating badge

4. Technical Integration Checklist (must align with current architecture)

- **Persona** → Update advisor persona JSON + RAG binding (see Collections section in screenshot). Primary collection remains `options-strategy` .
- **Templates** → Extend Templates UI logic (Next.js) to recognize the 5 new strategy cards

and route to the guided slot collector.

- **xChat Ask Flow** → Enhance `POST /api/xchat/ask` + NL preflight to detect template selection and spawn the preference-capture loop (server-driven, not pure client-side).
- **Async Job** → Leverage existing `strategy_jobs` + multi-agent handoff (`slots_complete`). Make the entire portfolio strategy build **asynchronous batch** (no blocking the chat thread).
- **DB / Preferences** → Extend or use `options_strategy_preferences` + new per-account fields. Ensure audit lineage.
- **RAG / Tools** → Confirm `options-coreskills` and strategy narratives are bound to the advisor persona.
- **UI Polish** → Dark-mode consistent, mobile-safe, HNWI elegance. Use existing Tailwind tokens and workspace rail branding.
- **Testing** → Update OpenAPI parity, add integration tests for template → preference capture → job trigger flow. Smoke test with real portfolio data.
- **Docs** → Update `xchat-personas.md`, `atx-multi-agent.md`, `PLAN.md` (add this as shipped feature), `current-state-features.md`.

Deliver this as a **production-ready, zero-downtime** change. Follow the exact PR hygiene in `.cursor/agents/reviewer.md` and `test-commit-push/SKILL.md`. Seed any new template narratives into `atx-docs/rag-collection/options-strategy/` if needed.

This feature will immediately elevate aTx Finance from a chat interface to a true **HNWI options strategy co-pilot** — conservative income machines, balanced range trades, and aggressive growth overlays, all built on the user's actual portfolios.

Let me know when you have the first PR ready for review. I will personally validate the strategy logic against live option chains and portfolio data.

— The Architect

Senior Finance Advisor & Options Trading Expert

aTx Finance – Production Systems for High-Net-Worth Portfolios